

Akhil S

Computer Science Undergraduate — AI/ML Engineer & Full-Stack Developer

akhilsenthil696@gmail.com LinkedIn GitHub Portfolio

Professional Summary

AI/ML-focused Computer Science undergraduate with hands-on experience in computer vision, NLP, LLM applications, and full-stack development. Built production-style systems using FastAPI, React, Docker, and vector databases, with research experience in applied AI and data-driven engineering solutions.

Education

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Amrita Vishwa Vidyapeetham, Coimbatore Campus

CGPA: 8.3/10

Experience

Python Development Intern (Virtual)

May 2025

ShadowFox

- Built Python-based solutions emphasizing clean code, modular design, and problem-solving.
- Collected, cleaned, and preprocessed datasets to improve data quality and support analytical workflows.

Projects

Centrifugal Atomizer Droplet Analysis using AI

- Processed 100k+ droplet images using OpenCV-based preprocessing and contour analysis.
- Developed CNN regression models for droplet size prediction with grouped cross-validation.
- Applied SAM segmentation and SHAP explainability to improve model interpretability.

RAPID-100 – AI-Powered Emergency Call Triage System

- Built real-time emergency call processing using speech-to-text, NLP, and ML classification.
- Implemented severity scoring, entity extraction, and dispatch decision support.
- Developed an end-to-end pipeline with FastAPI backend and real-time dashboard.

API Documentation & Testing Platform

- Developed a full-stack platform enabling developers to document, test, and monitor REST APIs through a unified interface.
- Implemented interactive endpoint exploration, automated request validation, load testing, and response-time analytics.

Evo Academy – AI-Powered Adaptive Learning Platform

- Built a personalized learning platform using LLMs and RAG for contextual Q&A and quizzes.
- Converted PDFs and notes into vector embeddings for semantic retrieval.
- Developed a scalable FastAPI, React/Next.js, and vector database architecture.

Research & Publications

Multidimensional Assessment of Drinking Water Scarcity

ICSRF 2025

- Conducted interdisciplinary research on socio-economic impacts of drinking water scarcity in Tamil Nadu using HCD and PRA methodologies.
- Performed stakeholder-focused analysis involving sustainability, accessibility, and resource challenges.
- Authored and presented findings at the International Conference on Sustainable Resilience (ICSRF 2025).

Skills

Programming: Python, Java, C, Embedded C, JavaScript, TypeScript, Shell Script

AI / ML: Computer Vision, NLP, CNNs, RAG, LLM APIs, Explainable AI (SHAP)

Web / Backend: React, Next.js, Node.js, Express, FastAPI

Databases: Weaviate DB, MongoDB, PostgreSQL, Firebase, Milvus

Tools: Git, Docker, Postman, OpenCV